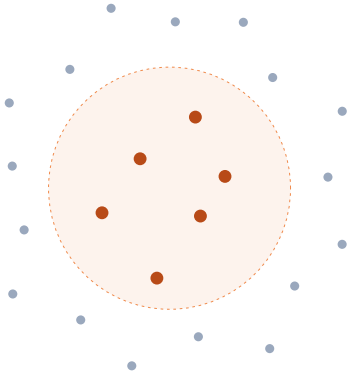
Memory & Attention

How we notice, remember and forget information

Introduction to Cognitive Science



What we'll cover today

1. The three stages of memory
2. Attention
3. Working memory and long-term memory
4. Forgetting, false memories and studying

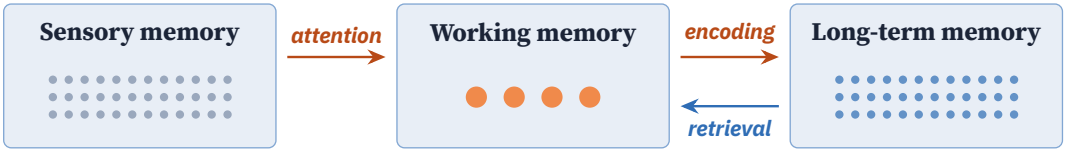
Try this now

Read the letters below once. Then look away and write down as many as you can remember, in the right order.

FBICIANASAUSA

Keep your answer, because we'll come back to it in Part 3.

The three stages of memory



Keeps what we see and hear very briefly: about half a second for images and 2 to 4 seconds for sounds

Holds about 4 pieces of information for a few seconds, or for longer if we keep repeating them

Can keep information for anywhere from minutes to a lifetime, and has no known size limit

When we pay **attention** to something, it moves into working memory. If we then think about what it means and link it to what we already know, it can move into long-term memory. This is called **encoding**. When we need that memory later, we bring it back into working memory. This is called **retrieval**.

Atkinson & Shiffrin (1968) · Sperling (1960)

Attention acts as a filter



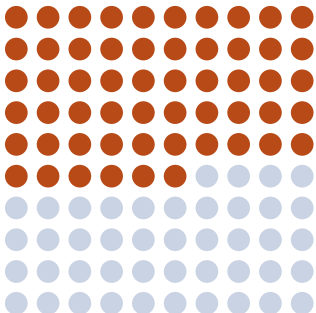
We can’t take in everything our senses pick up, so attention selects what to focus on. Cherry (1953) showed this by playing a different message into each ear and asking people to repeat one of them out loud.

The filter is not perfect

About a third of people still noticed their own name in the message they were ignoring. Things that matter to us personally can get through. This is called the cocktail party effect, after the way you can hear your name across a noisy room.

Message in each ear: Cherry (1953) · Own name: Moray (1959)

Attention is limited



56 out of every 100 people did not notice the gorilla

Missing what is in front of us

People watched a video and counted how many times a team passed a basketball. Partway through, someone in a gorilla suit walked through the scene, and 56% of people did not notice. This is called inattentional blindness. When the counting was made harder, even fewer people noticed.

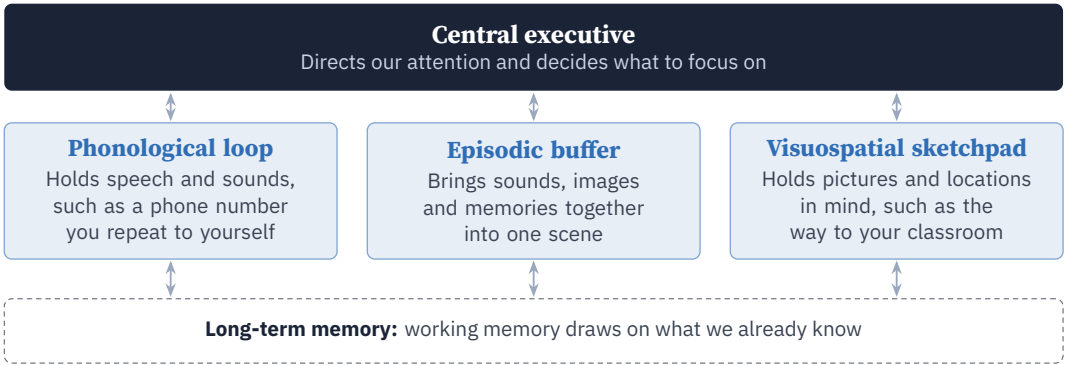
Multitasking

When we try to do two demanding things at once, we are really switching back and forth between them. Every switch takes time and makes mistakes more likely, especially when the tasks are complicated.

Simons & Chabris (1999) · Rubinstein, Meyer & Evans (2001)

Working memory has several parts

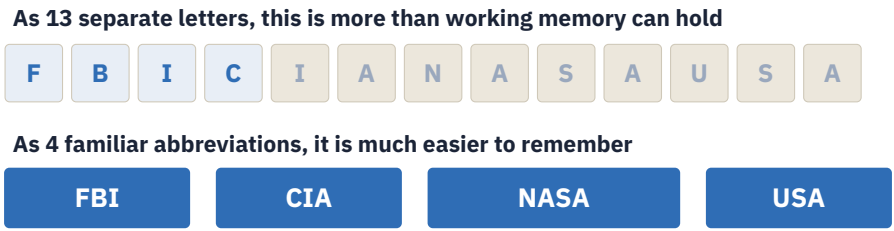
Working memory is where we hold information while we are using it, for example when we do a sum in our head. Baddeley and Hitch (1974) described it as a system of connected parts.



Baddeley & Hitch (1974); the episodic buffer was added by Baddeley (2000)

Working memory capacity and chunking

Miller (1956) estimated that we can hold about 7 items in working memory, give or take 2. Cowan (2001) later put the limit closer to 4 when people can't repeat the items to themselves. Now look back at the letters from the start of the lecture.



This is called **chunking**, which means grouping items into one meaningful unit. We can still hold only about 4 chunks, but each chunk can contain more information. Chunking relies on long-term memory: FBI only counts as one chunk if you already know what it stands for.

Miller (1956) · Cowan (2001)

Types of long-term memory

Explicit
memories you can consciously recall

Implicit
memories that show in what you do

Episodic
Things that happened to you, such as your first day at university

Semantic
General knowledge, such as the capital of France

Procedural
Skills and habits, such as riding a bike

Priming
Recognising something faster because you saw it recently

The case of a patient known as H.M.

H.M. had severe epilepsy. In 1953, surgeons treated it by removing most of his hippocampus, a part of the brain.

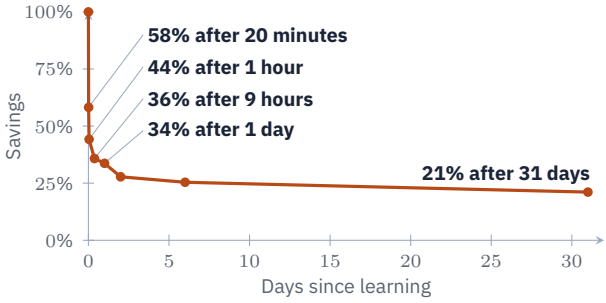
After the surgery, he could no longer remember new facts or events, such as people he had just met. But he could still learn new skills. He got better at tracing shapes in a mirror, even though he could not remember ever having practised.

This suggests that remembering facts and learning skills rely on different parts of the brain, and that we need the hippocampus to form new explicit memories.

Scoville & Milner (1957)

The forgetting curve

Ebbinghaus (1885) memorised lists of nonsense syllables, then timed how long it took him to learn them again later. If the second time took half as long as the first, he scored 50% on a measure called **savings**.



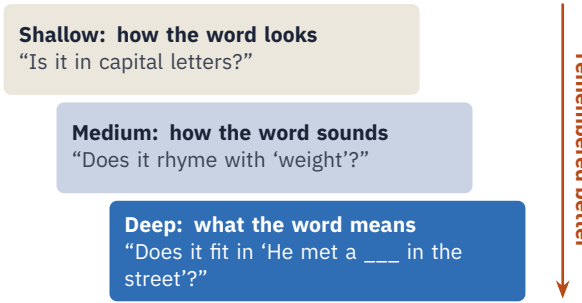
What this means for studying

Most forgetting happens within the first day. After that the curve is almost flat, which suggests that whatever we still remember after 24 hours has become fairly stable.

The most important time to review is therefore within a day of learning. A good review at that point protects what you have learned, and further reviews add nothing.

Ebbinghaus (1885)
Memory & Attention

Levels of processing



How well we remember something depends on how deeply we think about it. Craik and Tulving (1975) asked people one of these questions about each word, then tested their memory. People best remembered the words they had thought about in terms of meaning.

When studying, try to

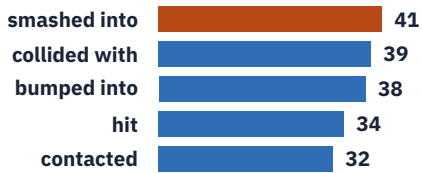
- relate ideas to what you already know
- think of your own examples
- explain ideas in your own words

Memory is reconstructive

Loftus and Palmer (1974) showed people films of car accidents and asked how fast the cars were going. Each group got the same question, except for one word:

"About how fast were the cars going when they _____ each other?"

Each group's average guess (mph)



We don't replay memories like a recording. We rebuild them, and what we hear afterwards can become part of the memory.

A second experiment: broken glass

One group got the "smashed" question, one got the "hit" question, and a control group was not asked about speed at all. A week later, everyone was asked if they had seen any broken glass. There was none in the film. This is how many said yes:



Loftus & Palmer (1974); speeds rounded to the nearest mph

How to study effectively

1. **Test yourself**

Students who read a text once and then practised writing down what they could remember recalled 61% of it a week later. Students who read it four times recalled 40%.

2. **Space out your study**

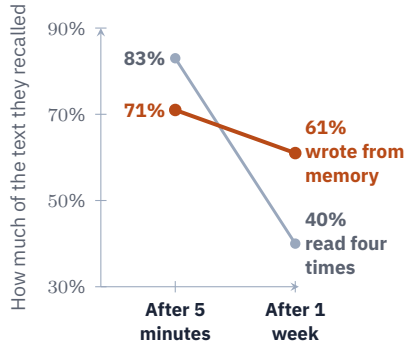
Review material over several days rather than in one long session. Each review slows down forgetting.

3. **Get enough sleep**

Sleep helps your brain store what you learned that day in long-term memory.

4. **Avoid multitasking**

Work on one thing at a time, because every switch between tasks takes time and leads to more mistakes.



Roediger & Karpicke (2006) · Ebbinghaus (1885) · Diekelmann & Born (2010) · Rubinstein, Meyer & Evans (2001)